

$$5. \begin{bmatrix} 5 & 1 & -8 & 4 \\ -2 & -7 & 3 & -5 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -8 \\ 16 \end{bmatrix}$$

From the definition of Ax , $5 \begin{bmatrix} 5 \\ -2 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ -7 \end{bmatrix} + 3 \begin{bmatrix} -8 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ -5 \end{bmatrix} = \begin{bmatrix} -8 \\ 16 \end{bmatrix}$

$$7. x_1 \begin{bmatrix} 4 \\ -1 \\ 7 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 3 \\ -3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -8 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$$

From the definition of Ax , $\begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -3 & 0 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$

In Exercises 9 and 10, write the system first as a vector equation and then as a matrix equation.

9. $3x_1 + x_2 - 5x_3 = 9$
 $x_2 + 4x_3 = 0$

Vector equation: $x_1 \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix}$

Matrix equation (from defn. of Ax)

$$\Rightarrow \begin{bmatrix} 3 & 1 & -5 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix}$$

10. $8x_1 - x_2 = 4$
 $5x_1 + 4x_2 = 1$
 $x_1 - 3x_2 = 2$

Vector equation: $x_1 \begin{bmatrix} 8 \\ 5 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

Matrix equation (from defn. of Ax)

$$\begin{bmatrix} 8 & -1 \\ 5 & 4 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

Given A and b in Exercises 11 and 12, write the augmented matrix for the linear system that corresponds to the matrix equation $Ax = b$. Then solve the system and write the solution as a vector.